GENZILLA V3.0 MULTI-TENANT SAAS

Store Admin System
Feature & Technical Documentation

Complete Architecture, Workflows, Subsystems & Regulatory Compliance Guide for Pharmacy Merchants

System Version:	**v3.0 Production Ready**
Target Domain:	**Health-Tech, Digital Pharmacy & Retail SaaS**
Architecture:	**Multi-Tenant Tenant-Aware Scoping**
Framework:	**Laravel 12 / PHP 8.3 / Tailwind CSS / Spatie RBAC**
Document Status:	**Official System Release Documentation**
Date:	**November 2026**

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Complete functional module map of the Genzilla Store Admin workspace

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17	**Database Schema & Architecture Cross-Reference** Model relationships, compound indices, cascading keys, and route map.	ENGINEERING

1. Executive Summary & Architectural Philosophy

The **Store Admin Workspace** of the Genzilla v3 Multi-Tenant SaaS platform is an enterprise-grade operational interface built specifically for pharmacy retailers, chain drugstores, and specialized medical supply merchants. Unlike standard retail software, modern digital pharmacies operate under strict drug regulatory guidelines, perishability schedules, and unique patient consultation demands.

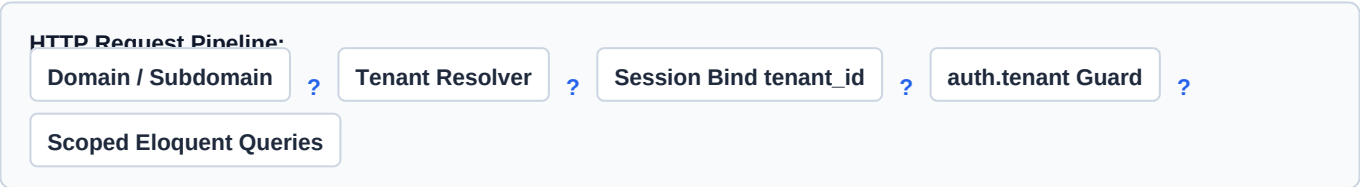
> **Core Mission of the Platform**
>
> Deliver an autonomous, compliant, and highly scalable cloud management portal empowering pharmacy staff to validate prescriptions, automate inventory depletion via FEFO, recover abandoned customer carts via tele-sales, and reconcile financial accounts across diverse payment gateways.

Key Differentiators in the Genzilla Store Architecture

- **Strict Prescription Compliance:** Regulated drugs require digital prescription upload and licensed pharmacist sign-off before dispatch, protecting merchants against regulatory sanctions.
- **FEFO Lot-Level Traceability:** Medicine inventory is tracked via distinct manufactured batches with assigned lot numbers and expiration dates, enforcing First-Expired, First-Out depletion.
- **Tele-Sales Recovery Pipeline:** Transforms abandoned checkouts into high-conversion tele-sales leads, allowing pharmacy staff to assist patients who lack immediate digital prescriptions.
- **Tamper-Proof Audit Trails:** Double-entry stock movements and transaction ledgers record every addition, deduction, reservation, and financial transfer.

2. Multi-Tenancy & Data Isolation Model

Genzilla v3 employs a **Tenant-Aware Scoping Model** within a shared database architecture. Each merchant operates in complete data isolation governed by row-level scoping, session container bindings, and compound database unique keys.



Tenant Isolation Guarantees

LAYER	ENFORCEMENT MECHANISM
Database Keys	All operational tables (`orders` , `products` , `inventory_batches` , `tenant_prescriptions` , `carts`) maintain a mandatory cascading foreign key referencing `tenants.id` .
Compound Uniqueness	SKUs, URL slugs, coupon codes, and order numbers enforce uniqueness per tenant (e.g., `UNIQUE (tenant_id, sku)`), allowing multiple merchants to utilize identical manufacturer codes safely.

LAYER	ENFORCEMENT MECHANISM
Session & Context	Resolved via the `tenantId()` helper function and registered globally in Laravel's service container during request bootstrapping.
Cross-Tenant Guard	Controllers execute strict ownership checks (e.g., `where('tenant_id', tenantId())->findOrFail($id)`), returning HTTP 404/403 upon any unauthorized access attempts.

3. Security, Authentication & Role-Based Access Control (RBAC)

The Store Admin portal uses **Spatie Laravel-Permission** configured with multi-tenant team scoping (`PermissionRegistrar::setPermissionsTeamId($tenantId)`). Store owners can configure granular roles and delegate daily operations to specialized staff members without compromising store governance.

Default Role Profiles

ROLE TITLE	PERMISSIONS SCOPE	SYSTEM SAFEGUARDS
STORE OWNER	Full access to all 35+ Store permissions including billing, settings, staff provisioning, and financial exports.	Permanent, protected account. Cannot be deleted or deactivated by any admin.
CHIEF PHARMACIST	Prescription review, order approval, catalog management, inventory receiving, batch adjustments.	Restricted from viewing platform subscription billing and deleting tenant admin accounts.
ORDER DISPATCHER	Order listing, order status transitions (Processing ? Shipped), invoice printing.	Read-only on customer database and catalog pricing configurations.
TELE-SALES REP	Recoverable carts, customer calling records, manual order entry, phone quotes.	Cannot alter store payment gateways or modify roles and permissions.

4. Store Admin Dashboard & Real-Time Analytics

The Dashboard (`/store/dashboard`) serves as the daily operational cockpit for store personnel. Built with lightweight Tailwind components and Chart.js, it updates instantly upon any transactional activity.

Dashboard Architecture & Live Widgets

- **Primary Operational KPIs:**
  - **Gross Sales (?):** Aggregates all non-cancelled orders with daily delta comparisons.
  - **Total Orders:** Real-time count of total orders, today's intake, and pending queues.
  - **Rx Pending Review:** Highlighted callout badge showing uploads awaiting pharmacist validation.
  - **Recoverable Carts:** Registered shoppers who abandoned items > 2 hours ago.
- **14-Day Dual-Axis Trends Chart:** Combined linear revenue plot (in BDT) alongside vertical bar graphs representing daily order intake.
- **Interactive Distribution Card:** Tabbed doughnut visualization allowing instant toggling between *Payment Preferences* (bKash, Nagad, COD, Card) and *Order Fulfillment Pipeline* (Delivered, Processing, Pending, Shipped, Cancelled).
- **Critical Stock Watchlist:** Automatically surfaces inventory items with available stock ? 10 units with SKU, reserved stock, and status warnings.
- **Recent Activity Feeds:** Live tables for incoming orders, pending prescriptions, and customer reviews moderation.

5. Prescription Management & Tele-Pharmacy Workflow

Digital pharmacies must balance user convenience with strict regulatory standards. The Genzilla prescription workflow bridges customer photo uploads with pharmacist verification and order fulfillment.

Prescription Lifecycle:
Patient Upload ? **Pharmacist Inspection** ? **Rx Order Conversion** ? **Dosage Note Builder** ? **Customer Payment Link**

Prescription Workspace Capabilities

FEATURE	DESCRIPTION & REGULATORY IMPACT
High-Res Image Viewer	Inspect doctor signatures, clinic stamps, patient details, and prescription issue dates in full resolution.
Order Conversion Engine	Pharmacists select medicines directly from catalog inventory, allocating specific batch lots and specifying custom dosage directions (e.g., *"1 tablet morning & night after food for 7 days"*).
Audit Logging	Stores `reviewed_by_user_id` and `reviewed_at` timestamps to provide complete regulatory accountability.
Rejection Management	Allows pharmacists to reject unreadable or expired prescriptions with formal rejection reason notifications dispatched to the patient.

6. Order Management, Checkout & Invoicing Pipeline

The order fulfillment pipeline coordinates inventory deductions, courier status updates, and automated customer notifications across both digital and manual telephone orders.

Fulfillment Lifecycle States

STATUS	OPERATIONAL DEFINITION	INVENTORY IMPACT
PENDING	Order received from storefront or prescription conversion. Awaiting staff review.	Inventory units flagged as `reserved`.
PROCESSING	Pharmacist confirmed order. Items are picked, verified, and packed.	Units remain reserved from designated batches.
SHIPPED	Parcel handed over to courier service with tracking assignment.	Units permanently deducted from `quantity`; reservations cleared.
DELIVERED	Customer received package and COD payment settled (if applicable).	Order finalized; financial transaction locked.
CANCELLED	Order revoked by customer or rejected due to stock/verification issues.	Reserved units released back to available batch stock.

7. Abandoned Checkouts & Recoverable Carts System

In online pharmacy retail, customers frequently abandon shopping carts because they do not have a prescription photo on hand, are uncertain about dosage pack sizes, or face unexpected delivery fees. The Genzilla system divides cart abandonment into two dedicated operational workflows.

Dual-Tab Operational Matrix

DIMENSION	TAB 1: RECOVERABLE CARTS (TELE-SALES)	TAB 2: TOP ABANDONED PRODUCTS (MERCHANDISING)
Primary Objective	Contact known patients and close pending sales via support.	Identify macro pricing, packaging, or stock frictions.
Target Population	Registered customers with known phone & email.	All users (Both registered customers & anonymous guests).
Data Granularity	1 Row per Cart / Shopper.	1 Row per Product Variant.
Key Data Points	Patient Name, Phone, Cart Items, Cart Total, Idle Time.	Variant SKU, Abandoned Count, Stuck Units, Lost Revenue.
Follow-Up Tracking	`recovery_contacted_at` timestamp and `recovery_note`.	Aggregated analytical trend.
Action Trigger	Direct WhatsApp message or phone call offering assistance.	Price adjustment, minimum delivery threshold reduction.

8. Transactions & Financial Payment Ledger

The Transactions module (`/store/transactions`) acts as a centralized double-entry financial ledger recording all customer payments, refunds, and gateway settlements.

Payment Reconciliations & Controls

- **Supported Payment Methods:** bKash Merchant API, Nagad Gateway, Rocket, SSLCommerz (Visa/Mastercard), Cash on Delivery (COD), and Direct Bank Transfer.
- **Manual Transaction Logging:** Staff can record offline customer cash/bank payments directly against open orders with transaction references and receipt uploads.
- **Refund Workflow:** Supports full and partial refunds with mandatory reason tracking. Reconciles order payment statuses automatically (e.g., transitions order to `partially_paid` or `refunded`).
- **Ledger Export:** Comprehensive CSV exports containing timestamps, gateway transaction IDs, customer details, and net settlement sums.

9. Product Catalog & Medicine Classifications

The Catalog architecture provides specialized fields required for pharmaceutical inventory management alongside standard retail attributes.

Catalog Data Structure

ATTRIBUTE GROUP	DATABASE FIELD	DESCRIPTION & USAGE
Medicine Data	`kind`	Enum (`medicine` or `general`) controlling regulatory UI behavior.
Generic Chemical	`generic_name`	Active pharmaceutical ingredient (e.g., *Paracetamol*, *Esomeprazole*) for generic substitution.
Legal Status	`requires_prescription`	Boolean flag enforcing mandatory doctor prescription upload at checkout.
Variant Matrix	`options_hash`	MD5 attribute combination hash supporting multiple dosage/strip sizes (10s Strip, 100s Box).
SEO Metadata	`seo_title` , `seo_description`	Configurable search engine tags for organic Google rankings.

10. FEFO Batch Inventory & Stock Movement Ledger

To eliminate expired medicine losses and ensure patient safety, Genzilla utilizes **First-Expired, First-Out (FEFO)** batch allocation paired with an immutable double-entry stock movement ledger.

> **Regulatory Rule: Lot-Level Traceability**
>
> Every physical medicine unit in the warehouse must be associated with a valid manufacturer batch number, cost price, and expiration date. Stock quantities are never mutated without writing a signed entry to the `stock_movements` table.

Stock Movement Audit Types

MOVEMENT TYPE	TRIGGER EVENT	LEDGER ENTRY IMPACT
`in`	Receiving shipment from pharmaceutical distributor.	Increments batch quantity; logs batch #, expiry date, unit cost price.
`reserved`	Customer places order on storefront or phone desk.	Increments batch `reserved` count; prevents overselling.
`out`	Order dispatched and marked as Shipped.	Decrements both `quantity` and `reserved` ; writes order ref.
`released`	Order cancelled or payment expired.	Decrements `reserved` count; restores available stock.
`adjustment`	Physical stock audit, damaged goods, or laboratory quarantine.	Adjusts batch quantity with required manager explanation.

11. Product Reviews & Customer Reputation System

The Product Reviews moderation workspace (`/store/catalog/reviews`) allows store administrators to protect store credibility while fostering transparent patient feedback.

- **Editorial Moderation:** All customer reviews enter as `pending` and require explicit approval before appearing on the public storefront.
- **Verified Pharmacist Replies:** Staff can author public responses to customer questions or adverse reaction reports, showing professional medical care.
- **Bulk Actions:** Batch approve or delete spam reviews with single-click bulk tools.

12. Customer Relationship Management (CRM) & Directory

The Customer Directory (`/store/customers`) provides a unified view of patient histories, order frequencies, and address books.

- **Patient Lifetime Value (LTV):** Displays total orders placed, cumulative turnover, and average order value.
- **Multi-Address Support:** Patients can store multiple delivery addresses (Home, Office, Hospital Ward) with designated defaults.
- **Account Safeguards:** Quick status toggles (`active` , `suspended` , `inactive`) prevent repeat fraudulent checkouts.

13. Promotions, Dynamic Coupons & Delivery Charges

Merchants can craft targeted promotional incentives while setting strict margin guardrails.

Coupons & Geo-Targeted Delivery Charges

MODULE	CAPABILITIES	BUSINESS SAFEGUARDS
Coupons	Percentage discounts (e.g. 10%) or fixed amounts (e.g. ? 100).	Minimum cart spend thresholds, maximum total usage, and per-user limits.
Delivery Charges	Geo-zoned tiers (Inside Dhaka: ? 60, Outside Dhaka: ? 120, Express Delivery: ? 150).	Automated fee waivers when customer subtotal surpasses designated thresholds.

14. Storefront CMS & 15. Store Settings

Store owners maintain complete control over visual marketing assets, custom domain routing, and platform billing without engineering intervention.

- **Storefront CMS:** Manage hero sliders, promotional banner grids, and policy pages (Terms of Service, Privacy Policy, Medicine Return Guidelines).
- **Store Settings:** Store branding, pharmacy license numbers, drug registration identifiers, and hotline contact channels.
- **Custom Domains:** Self-service DNS mapping and automated CNAME/A-record verification.
- **SaaS Subscription:** Plan quota monitoring (products, orders, storage limits) and invoice receipt downloads.

16. Business Intelligence & Reporting Suite

The reporting suite (`/store/reports`) equips merchants with actionable intelligence across four critical operational dimensions:

1. **Sales Performance:** Daily/monthly gross revenue, discounts given, delivery fees, and net settlement totals.
2. **Inventory Valuation & Expiry Audit:** Total warehouse valuation, units on hand, and medicines expiring within 30, 60, or 90 days.
3. **Order Fulfillment Report:** Timeliness metrics, cancellation causes, and return ratios.
4. **Customer Acquisition:** Patient signup growth, repeat purchase rates, and top spenders.

17. Database Schema & Architecture Cross-Reference

MODEL	DATABASE TABLE	KEY RESPONSIBILITIES
`Tenant`	`tenants`	Tenant root record, store name, subdomain, subscription status.
`User`	`users`	Store admins and pharmacy staff; authenticated via `auth.tenant` .
`TenantPrescription`	`tenant_prescriptions`	Prescription photo uploads, patient notes, conversion status.
`Order`	`orders`	Fulfillment record, delivery address, subtotal, discount, net total.
`OrderProduct`	`order_products`	Line items with unit price, quantity, batch allocation, Rx flag.
`OrderPayment`	`order_payments`	Transaction log tracking gateway ID, amount, and payment status.
`Cart` / `CartItem`	`carts` / `cart_items`	Shopping carts, guest session tokens, recovery notes & timestamps.
`Product` / `Variant`	`products` / `product_variants`	Catalog definitions, generic drug names, SKUs, and prices.

MODEL	DATABASE TABLE	KEY RESPONSIBILITIES
`InventoryBatch`	`inventory_batches`	Physical lot numbers, expiration dates, cost accounting, available stock.
`StockMovement`	`stock_movements`	Tamper-proof double-entry inventory transaction log.
`ProductReview`	`product_reviews`	Customer star ratings, moderation state, and official staff replies.
